

PYTHON PROGRAMMING – IMPORTANT TOPICS & QUESTIONS

UNIT I – Basics of Python

Important Topics

- History & Features of Python
- Tokens & Character Set
- Core Data Types (int, float, str, bool)
- Input/Output functions
- Type conversion
- Operators (Arithmetic, Bitwise, Assignment)
- Operator precedence
- Formatting strings
- Built-in functions

Important Questions

2–5 Marks:

1. Explain Python tokens.
2. List Python core data types.
3. Define operator precedence.
4. What are compound assignment operators?
5. Write a program to format numbers and strings.

10–16 Marks:

1. Explain Python operators with examples.
2. Write a program using arithmetic and bitwise operators.
3. Explain internal working of Python.
4. Write simple programs using I/O and formatting.

Most Important Topics:

- ✓ Operators & Expressions
- ✓ Core Data Types
- ✓ Built-in Functions

UNIT II – Control Statements & Functions

★ Important Topics

- Boolean type & relational operators
- Decision statements (if, if-else, nested if)
- Loops (while, for)
- break & continue
- range() function
- Functions
- Parameters & Arguments
- Scope (Local & Global)
- Recursion
- Lambda functions

Important Questions

Short Questions:

1. Difference between while and for loop.
2. What is recursion?
3. Define lambda function.
4. Explain scope of variables.
5. What is range() function?

Long Questions:

1. Explain decision statements with example.
2. Write a program using nested loops.
3. Explain functions with parameters and return statement.
4. Explain recursion with example program.

Very Important:

- ✓ Functions & Recursion
- ✓ Scope of Variables
- ✓ Loop Control Statements

✅ UNIT III – Strings, Lists, Tuples, Sets, Dictionaries

★ Important Topics

- String operations

- String immutability
- List creation & slicing
- List comprehensions
- List methods
- Tuples
- Sets
- Dictionaries

Important Questions

Short Questions:

1. What is string immutability?
2. Define list slicing.
3. What are tuples?
4. Difference between list and tuple.
5. Define dictionary.

Long Questions:

1. Explain string operations with examples.
2. Explain list methods and list comprehensions.
3. Compare List, Tuple, Set, Dictionary.
4. Write programs using dictionary and set operations.

Highly Important:

- ✓ List & List Comprehension
- ✓ String Operations
- ✓ Difference between List, Tuple, Set, Dictionary

UNIT IV – Exception Handling, File Handling & Modules

Important Topics

- Types of Exceptions
- try, except, finally
- File Handling (read, write, append)
- seek() function
- Binary files

- Modules & Packages
- Importing modules

Important Questions

Short Questions:

1. What is exception handling?
2. Define seek() function.
3. What is a module?
4. Difference between text and binary file.

Long Questions:

1. Explain Python exception handling with example.
2. Explain file handling operations.
3. Explain modules and packages in Python.

Most Important:

- ✓ Exception Handling
 - ✓ File Handling
 - ✓ Modules & Packages
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UNIT V – Object Oriented Programming & Turtle

Important Topics

- Class & Object
- Constructor (**init**)
- Destructor (**del**)
- Instance & Class variables
- Method overloading
- Operator overloading
- Inheritance
- Encapsulation
- Turtle module

Important Questions

Short Questions:

1. Define class and object.

2. What is constructor?
3. What is inheritance?
4. Define encapsulation.
5. What is turtle module?

Long Questions:

1. Explain OOP concepts in Python.
2. Write a program using class and constructor.
3. Explain inheritance with example.
4. Explain operator overloading.
5. Write a simple turtle graphics program.

Very Important:

- ✓ OOP Concepts
- ✓ Constructor & Inheritance
- ✓ Turtle Graphics Program

HIGH PRIORITY TOPICS (Study First)

If exam is near, focus on:

1. Functions & Recursion
2. Lists & List Comprehension
3. File Handling
4. Exception Handling
5. OOP Concepts
6. Inheritance
7. Decision & Loop Statements

Unit Weightage Suggestion

Unit	Difficulty	Exam Weight
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Unit I	Easy	Medium
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Unit II	Moderate	High
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Unit III	Moderate	High
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Unit	Difficulty	Exam Weight
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Unit IV	Moderate	Medium
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Unit V	Moderate	High
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